

CODEQUEST 2026 · ROUND 2 · PROTOTYPE

LoanWizard

Apply for a loan by talking to your camera. The model watches, listens, verifies, and prices the offer in minutes, not days.

AI video loan origination for Indian digital lending

Rishet Mehra · github.com/Rishet11/LoanWizard

THE PROBLEM

Loan origination in India still runs on paper and patience.

SLOW

Days, not minutes

Branch visits, photocopied documents, hand-keyed forms, multi-day waits before a decision.

OPAQUE

A black box

The applicant is told yes or no with no reason they can see, and no record the lender can defend.

BOLTED ON

Video as theatre

Video-KYC made remote onboarding legal, but most products bolt a call onto the same old paper process.

The decision stays a black box, and the audit trail is thin. That is the gap.

THE SOLUTION

The interview *is* the application.

SPEAK

The form fills itself

Multilingual speech-to-text extracts name, employment, income, amount and purpose as you talk.

VERIFY

The camera checks you live

Passive liveness — blink, head-pose, texture — plus on-device Aadhaar and PAN OCR. No uploads.

PRICE

Real-time decision

A risk MLP, a fraud MLP, a persona model and a policy engine produce a rate, an EMI and reason codes.

PROVE

Explainable & replayable

Every offer ships reason codes and a narrative; the input snapshot is frozen so it can be re-run.

One continuous experience — and compliance is first class, not an afterthought.

THE EXPERIENCE

From link to offer, in one sitting.

1

Operator Console

A single-screen lending terminal — live perception gauges, the video feed, one Initiate control.

2

Permission gate

Plain-language reasons before camera, mic and location are ever requested.

3

Video interview

The agent asks, the customer speaks, the form fills; liveness and fraud signals tick live in a co-pilot panel.

4

The offer

A card with rate, EMI, weighted reason codes and a plain-language narrated explanation.

5

Admin & audit

Session timeline, fraud and consent evidence, fairness and drift dashboards, one-click decision replay.

Three streams, one frozen contract.

STREAM A · BROWSER

Perception

Camera, mic, speech-to-text, TF.js liveness, document OCR, age estimate, device fingerprint. Emits typed events.

STREAM C · WEB

Next.js 14

Session state machine, Operator Console, Prisma audit writes, admin dashboards, i18n, SSE.

STREAM B · SERVICE

FastAPI

Risk MLP, fraud MLP, persona, policy engine, bureau merge, reason narrator, drift, fairness, replay.

Server-authoritative: the client never submits the offer inputs — the ML service recomputes from persisted session data. That is what makes the audit trail trustworthy. Full diagrams: [docs/architecture-current.pdf](#) & [architecture-v4.pdf](#)

Every offer runs the full chain.

1 Bureau merge
mock CIBIL + Experian behind an adapter

2 Policy engine
hard rules: age, income, cap, liveness, face

3 Risk MLP
Keras v1.2.0 → risk band

4 Fraud MLP
Keras v0.1.0 → fraud score

5 Persona classifier
rules v1.0.0 → applicant label

6 Offer builder
band + policy → amount, rate, tenure, EMI

7 Reason narrator
weighted codes + plain language

8 Audit + snapshot
frozen inputs, model versions, drift sample

Explanations derived from the model — not dressed up to look like it.

Permutation importance

- Baseline prediction, then perturb each of 12 features to its scaled mean, re-predict, measure the change.
- Runs as a single batched predict — cheap enough for every request.
- Reason codes reflect what the model **actually responds to**, not raw feature size.

Replay on the exact version

- Each decision is stamped with its model versions.
- Weights are archived per version; replay loads the model **as it was**.
- Reports `exact_model_match` — and flags it honestly on fallback.

Models are trained on synthetic data with scripts in the repo, so the committed weights are reproducible from scratch.

The same flow that prices the loan produces the evidence.

RBI Video-KYC

Live video interview, liveness checks, recording-retention notice.

DPDP Act 2023

Explicit consent, disclosed data use, a working right-to-forget endpoint.

Auditability

Frozen snapshots, per-decision model versions, one-click replay.

Fair lending

Cohort fairness report and feature-drift tracking in the admin console.

Borrower protection

3-day cool-off on the accepted screen, downloadable Key Fact Statement.

Tamper resistance

Offer computed server-side from persisted data — the client can't forge inputs.

What is real, and what is mocked.

Real & running

- Full ML pipeline: risk MLP, fraud MLP, persona, policy, narrator
- Permutation-importance reason codes
- Audit + frozen snapshots + replay on exact version
- Browser perception locally: liveness, OCR, age, speech
- Admin: sessions, fairness, drift, replay

Mocked / optional by design

- Bureau pull — adapter in place, no commercial credentials
- LLM narration — template default, Gemini optional
- Whisper STT model — opt-in deploy toggle (fallback wired)
- Public judge link — scripted perception + reliable mock offer

We would rather be precise than oversell. The real camera and ML path runs locally and the ML Space serves the live /offer, /docs and /fairness endpoints.

ENGINEERING

A small, honest, reproducible stack.

105

ML service tests passing

33 / 33

perception tests passing

0

secrets needed to build the web app

Next.js 14

FastAPI

TensorFlow / Keras

TensorFlow.js

scikit-learn

Tesseract.js

@vladmandic/face-api

PostgreSQL

Prisma

SQLAlchemy

Zod

pnpm + Turborepo

Docker

Hugging Face Spaces

Neon

Three parallel streams against a frozen contract package. ML service is stateless and scales horizontally; perception runs on the client.

ROADMAP

The contract boundary carries from MVP to platform.

NOW MVP Real decision pipeline, perception, audit, replay, admin.	v1 Real bureau + auth Live CIBIL/Experian behind the adapter; object store; OIDC.	v2 Event spine + serving Kafka, TF Serving, feature store, vector DB.	v3 LLM + fraud ML Gemma narration, graph fraud, active learning.	v4 Platform Multi-tenant, disbursement, servicing, co-lending.
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Near-term: deepfake-resistant liveness via the optional texture model, and a lender console for human-in-the-loop review on borderline decisions.

IN ONE LINE

A customer talks to their camera and walks away with an **explainable, auditable, replayable** loan offer — in minutes, not days.

Repository: github.com/Rishet11/LoanWizard

Docs: [README.md](#) · [QNA.md](#) · [docs/technical-documentation.pdf](#) · [docs/architecture-current.pdf](#)

Run it: `pnpm install · docker compose up -d · prisma db push · pnpm --filter @loan-wizard/web dev`